

BIRTH: Tracer-Aware Initial Segmentation and Stateless Topology-Safe Interaction for the autoPET V Challenge

Abstract. The autoPET V challenge evaluates interactive whole-body lesion segmentation in FDG- and PSMA-PET/CT using voxel-overlap and lesion-detection metrics over an initial prediction and five corrective interactions. We developed a tracer-aware pipeline that combines a strong five-fold initial segmentation with conservative, stateless click correction. A fixed ensemble router first identifies the tracer. The initial mask is generated by the AutoPET-III LesionTracer ensemble and is calibrated separately for FDG and PSMA. FDG uses a probability threshold of 0.47 and burden-aware connected-component filtering, while PSMA uses a frozen component-level false-positive classifier based on probability, uptake, size, and spatial features. Interactive calls reconstruct the initial mask and replay all cumulative scribbles through a fold-0 EDT correction model. Only click-targeted foreground components are accepted, lesion bridges are rejected, and background deletion is disabled to prevent unstable fragmentation. On the Preliminary Test Set, the submitted container achieved a Dice score of 0.853649 and a lesion F1 score of 0.822220, ranking third at submission time. In local patient-independent evaluation, the final FDG interaction policy increased AUC-Dice by 0.011923 and AUC-DMM by 0.033626. The complete source, container recipe, and frozen interactive weights are publicly available.

Keywords: PET/CT lesion segmentation · interactive segmentation · autoPET V · nnU-Net · lesion detection

1 Introduction

Whole-body PET/CT combines metabolic and anatomical information and is widely used for cancer diagnosis, staging, and therapy-response assessment [1, 2]. Reliable lesion delineation is nevertheless laborious because disease burden may comprise many small, spatially separated lesions and because physiological uptake can resemble malignancy. The autoPET challenge series has established a large multi-tracer benchmark for automated lesion segmentation and has highlighted the importance of generalization across centers, scanners, and tracer distributions [3].

autoPET V focuses on human-AI collaboration. Each algorithm first predicts a segmentation without guidance and subsequently receives corrective foreground or background scribbles. The official ranking integrates Dice and the

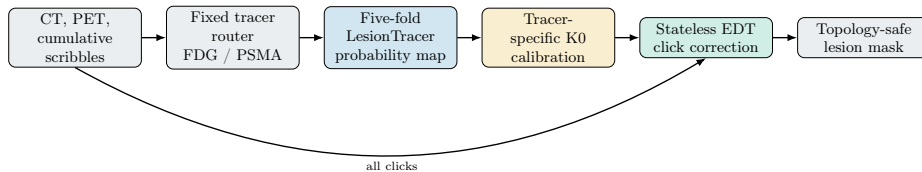


Fig. 1. Graphical overview of the final BIRTH submission. The zero-click prediction follows the upper path through tracer-specific calibration. When scribbles are present, the same K0 is reconstructed and cumulative clicks are replayed through a conservative EDT transaction.

Detection Matching Metric (DMM) across six states, consisting of K0 and five interactive corrections. Consequently, a competitive method must combine a strong initial segmentation with predictable click response. An aggressive interactive model can improve missed lesions but can also reduce area under the curve (AUC) if a correction changes distant anatomy or fragments a correct lesion.

Our final system therefore separates initial segmentation from interaction. The initial prediction uses a fixed five-fold lesion model followed by tracer-specific calibration. Interaction is stateless and component-local: the initial mask is regenerated for every invocation, and only the component targeted by cumulative foreground guidance may be added. This design preserves the high-quality baseline while providing a controlled path for correcting false negatives.

2 Methods

2.1 Data

Training data. We used the publicly released autoPET FDG and PSMA PET/CT data provided through the challenge series [4, 5]. The segmentation backbone was the publicly available five-fold AutoPET-III LesionTracer model. The interactive branch used the publicly released fold-0 AutoPET-IV LesionLocator EDT checkpoint. No hidden test image, Preliminary Test Set annotation, or external private dataset was used for training or parameter fitting.

Validation data. All calibration decisions were made on labeled challenge training cases using patient-independent partitions. Patient identity, rather than scan identity, was hashed into calibration and holdout partitions so that longitudinal scans of the same patient could not occur in both groups. K0 development used 206 FDG cases (105 lesion-present and 101 lesion-absent) and 118 PSMA cases (104 lesion-present and 14 lesion-absent). Interaction policies were evaluated over 105 lesion-present FDG and 104 lesion-present PSMA cases using six prediction states per case. Preliminary cases were used only as a final fail-closed regression check after all thresholds had been frozen.

2.2 Data pre-processing

Input MHA volumes are converted to NIfTI without changing physical geometry. The segmentation model applies its frozen nnU-Net preprocessing, including CT intensity normalization and PET normalization. A fixed three-classifier router uses robust whole-body PET/CT distribution features to select FDG or PSMA. All non-finite values are sanitized before routing. The output mask is converted back to MHA and copies the size, spacing, origin, and direction of the input CT.

2.3 Algorithm/model

Five-fold initial segmentation. The initial probability map is the mean output of five AutoPET-III LesionTracer folds using the residual-encoder nnU-Net architecture [6, 7]. Mirror test-time augmentation is disabled. The five-fold model is invoked in an isolated subprocess because its frozen nnU-Net fork is incompatible with the interactive fork. This isolation also releases its CUDA context before the EDT branch is initialized.

FDG K0 calibration. FDG probabilities are binarized at 0.47. A robust mask first removes connected components smaller than 25 voxels. If the robust mask contains at least 25 components, indicating high lesion burden, the minimum component size is relaxed to five voxels. This burden gate improved recall in high-load FDG studies while preserving conservative filtering in low-load and lesion-absent studies.

PSMA K0 calibration. PSMA probabilities are binarized at 0.50 and components smaller than five voxels are removed. A frozen logistic component classifier then estimates the probability that each predicted component is false positive. Its inputs are log-transformed voxel count and physical volume, probability maximum, mean and 90th percentile, PET SUV maximum and mean, and normalized three-dimensional centroid. Components with false-positive probability at least 0.86 are removed. The threshold was selected on the calibration partition and evaluated once on the patient-independent holdout partition.

Stateless interactive correction. The interaction branch uses a residual-encoder EDT model initialized from its frozen fold-0 checkpoint. Every container call recomputes K0 and constructs a prediction from K0 plus all cumulative clicks; no previous container state is required. For a foreground correction, connected components are extracted from the EDT prediction and only components intersecting a valid foreground point are proposed. A proposed union is rejected if it bridges two components that were separate in K0. Background edits are disabled because local validation showed that partial deletion could fragment correct lesions and increase instance-level false positives. For PSMA, foreground correction is further restricted to at least six cumulative tumor points and at most ten K0 components. Figure 2 illustrates the transaction.

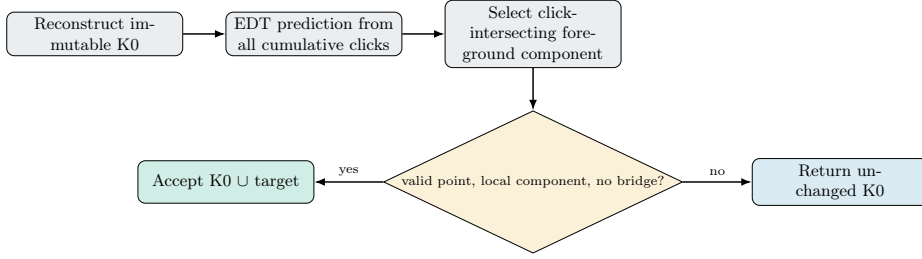


Fig. 2. Topology-safe stateless interaction. Background deletion is disabled; only a foreground component supported by the cumulative click set can modify the immutable K0 mask.

Table 1. Frozen inference parameters of the final submission.

Component	FDG	PSMA
Probability threshold	0.47	0.50
Minimum K0 component	25 voxels; 5 if ≥ 25 robust components	5 voxels
Learned FP pruning	none	logistic threshold 0.86
EDT folds	fold 0	fold 0
Minimum tumor points	1	6
Maximum K0 components	unrestricted	10
Background deletion	disabled	disabled

2.4 Data post-processing

All post-processing is part of the deterministic K0 procedure. FDG uses the 0.47 probability threshold and burden-aware 25/5-voxel dust rule. PSMA uses a 0.50 threshold, five-voxel dust, and the component classifier. No morphological closing, hole filling, or organ mask is applied after interaction. Final masks are binary unsigned 8-bit volumes in the CT reference geometry.

2.5 Training and test parameters

The submitted segmentation and interaction networks were frozen public checkpoints; no network parameter was updated during our final method development. Initial inference uses five folds, Gaussian sliding-window weighting, step size 0.5, and no mirroring. EDT uses fold 0, Gaussian weighting, step size 0.5, and no mirroring. K0 thresholds, component limits, and click gates are listed in Table 1. The container uses PyTorch 2.6, CUDA 12.4, NumPy 2.1.2, and SimpleITK 2.5.5.

Table 2. Local holdout effect of the tracer-specific K0 calibration. Values in parentheses are changes from the uncalibrated five-fold K0.

Tracer	Dice	Macro F1	Micro F1	TP/FP/FN	Main effect
FDG	0.895932 (−0.000021)	0.790105 (+0.000267)	0.722334 (+0.000726)	359/64/212	FP −1
PSMA	0.787735 (−0.000002)	0.866997 (+0.000571)	0.794293 (+0.001179)	1336/336/356	FP −5

2.6 GitHub repository

Source code, the reproducible Docker recipe, unit tests, and frozen interactive weights are available in our public GitHub repository under the project name `autoPET-V-BIRTH-final` (account: `hemingduo85-droid`). The AutoPET-III weights are downloaded from their original Zenodo record. All downloaded weights are checked by MD5 or SHA-256 during the Docker build.

3 Results

3.1 Initial prediction

The final container was evaluated on the Preliminary Test Set on 31 August 2026. It achieved Dice 0.853649 and lesion F1 0.822220 and was ranked third by the displayed mean-rank score. Relative to our preceding submission, both metrics increased. Table 2 summarizes patient-independent local holdout results for the tracer-specific K0 changes. FDG threshold calibration improved macro and micro F1 with a Dice change of -2.1×10^{-5} . PSMA component pruning retained TP and FN counts while reducing FP from 341 to 336.

3.2 Interactive evaluation

The interaction policy was evaluated with K0 and five simulated corrections. For the final FDG K0, mean AUC improvement relative to K0 was 0.011923 for Dice and 0.033626 for DMM over 105 lesion-present cases. A 10,000-sample case bootstrap yielded 95% confidence intervals of [0.001799, 0.029481] and [0.015610, 0.057647], respectively. Thirty-nine cases improved in AUC-Dice, 50 were unchanged, and 16 decreased; for AUC-DMM, 20 improved, 84 were unchanged, and one decreased.

In a 104-case PSMA study used to freeze the interaction gate, AUC-Dice and AUC-DMM increased by 0.014433 and 0.014108, respectively. The corresponding 95% bootstrap intervals were [0.00710, 0.02300] and [0.00607, 0.02380]. These results motivated the PSMA point-count and component-count restrictions used in the final container.

4 Discussion

The proposed pipeline prioritizes expected AUC rather than the largest possible single-step correction. The strong five-fold K0 limits the room for safe improvement, especially in high-burden studies where a foreground component can merge

nearby lesions. Component-local transactions and bridge rejection reduce this risk. Disabling background deletion also reflects a deliberate precision–stability trade-off: false-positive regions are not interactively removed, but correct lesions cannot be fragmented by a negative correction.

The principal limitation is that the interactive network uses a single fold and was not retrained jointly with the final K0 calibration. Moreover, the clinician-driven category uses a fixed set of expert scribbles collected with respect to the challenge baseline. Such scribbles may not coincide with errors of our K0, so performance in that category cannot be inferred perfectly from simulated interaction. Finally, the displayed Preliminary Test Set contains few cases and is not representative of the multi-center final cohort. The reported local improvements should therefore be interpreted as validation evidence rather than a guarantee of final rank.

5 Conclusion

We presented a reproducible, tracer-aware pipeline for interactive lesion segmentation in whole-body PET/CT. The method combines a high-quality five-fold initial prediction, conservative FDG and PSMA calibration, and stateless topology-safe EDT correction. The submitted K0 improved both Dice and lesion F1 on the Preliminary Test Set, while local six-step studies showed positive AUC-Dice and AUC-DMM improvements. The modular design and public container recipe support independent reproduction and further analysis.

Acknowledgments. This work was supported by Tsinghua University Initiative Scientific Research Program (Student Academic Research Advancement Program).

Disclosure of Interests. The authors declare that they have no competing interests relevant to this work.

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Table 3. Algorithm details required by the autoPET V method-description template.

Team name	Algorithm name	Data pre-processing	Data post-processing	Training augmentation
BIRTH Lab	AutoPET V Final	Frozen nnU-Net CT/PET normalization; robust tracer routing	FDG threshold and burden dust; PSMA component pruning	Frozen public checkpoints; no additional network training
Test-time augmentation	Ensembling	Standardized framework	Network architecture	Loss
None	Five-fold K0; fold-0 EDT	nnU-Net v2 forks	3D residual encoder U-Nets	Frozen checkpoints; original Dice/CE objectives
Training data	Model dimensionality	Pre-trained models		Hardware
Public autoPET FDG and PSMA PET/CT	3D whole-body sliding-window inference; K0 patch defined by frozen model	AutoPET-III LesionTracer and AutoPET-IV LesionLocator EDT		NVIDIA A40 for validation and packaging

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